

AMSS 2026 — Lecture 11: Model Evaluation & Quality

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Today's Agenda

1. Frame: from generating to judging
2. Three quality criteria; how to evaluate
3. Live demo: ask AI to evaluate a flawed artifact
4. Where AI is a worse evaluator than you
5. How to use AI as an evaluator — and how not to
6. Bridge to Week 12 + the project

Recap: Architect and Critic

- ▶ **Architect / director:** drive AI through the software development lifecycle (SDLC).
- ▶ **Critic / reviewer:** read AI's output and name what's wrong or missing.

For nine weeks **you** were the critic. Today's question: can **AI** be the critic — can it judge its own output?

From Generating to Judging

- ▶ **Weeks 2-10: AI generated, you judged.** Every read-order trained your judgement.
- ▶ **Today: what if AI judges?** It is fast and tireless — but is it *right*?

Today's question: **which parts of quality can AI evaluate — and which must stay yours?**

The Literacy Floor: Critique, Rationale, Traceability

From Week 1: in the oral defense, *unaided*, you must demonstrate:

- ▶ **Critique** — read & critique AI-generated artifacts on the spot.
- ▶ **Rationale** — articulate why you directed AI a certain way.
- ▶ **Traceability** — defend the trace across your project.

Critique is the human's evaluation skill. Today is why it can't be delegated — the defense is *unaided* for a reason.

Three Quality Criteria

- ▶ **Consistency** — no contradictions, within an artifact and across them.
- ▶ **Completeness** — covers every requirement and case (nothing missing).
- ▶ **Correctness** — matches the domain and reality (it is actually right).

The three C's. Every evaluation asks all three.

The Three Are Not Equal — For AI

- ▶ **Consistency** — largely *mechanical*: can be checked against rules.
- ▶ **Completeness** — needs *intent*: you must know what *should* be there.
- ▶ **Correctness** — needs *domain truth*: you must know what's actually right.

AI can assist with the first. The last two are **yours**.

How You Evaluate: Three Techniques

- ▶ **Static evaluation** — read against quality rules. *The Weeks 4-10 read-orders are exactly this.*
- ▶ **Conformance checking** — does the model obey its well-formedness rules?
- ▶ **Simulation** — execute the model: run the state machine, walk the sequence, run the test.

Brief: Metamodel + Conformance

- ▶ A **metamodel** is the rules for what a *valid model* is — UML's own model of itself.
- ▶ **Conformance** = your model obeys them: every association has two ends; a state machine has an initial state; every message has a sender and receiver.

A mechanical check — and exactly where AI **can** help.

Your Turn: Predict the Verdict

In pairs, write down whether AI will (a) praise the flawed bike-sharing diagram, (b) nitpick cosmetics, or (c) catch the real defects — then we reveal live.

Demo: Ask AI to Evaluate

Live: hand AI a flawed bike-sharing class diagram — one we know the defects of — and ask “is this good?”. Watch what it says.

Prompt to AI: *“Here is a UML class diagram for a bike-sharing app. Is this a good design? Rate it. [paste a flawed diagram]”*

AI Evaluates to Please — Not to Judge

AI is trained to be helpful and agreeable. Asked to judge, it leans toward approval. The critic question turns on AI itself:

“Is this judgement independent — or is AI just agreeing with me, or with itself?”

Failure #1: Sycophancy

“Is this design good?” -> “Yes — *this is a clean, well-structured design!*”

Frame it as yours (“I made this”) and the praise only grows.

Lesson: *AI rates to agree. A flattering evaluation is not a passing one.*

Failure #2: Plausibility Bias

AI judges whether the artifact **looks** right — professional names, tidy layout — not whether it **is** right against the domain.

A clean diagram with a wrong multiplicity passes.

Lesson: *Looks-right is not is-right. AI optimises for the first.*

Failure #3: Blind to Absence (Completeness)

AI evaluates **what is there**. It cannot reliably tell you **what is missing** — it doesn't know your full intent.

The dropped requirement, the unhandled failure path — invisible to it.

Lesson: *Completeness is yours. AI can't miss what it never knew to expect.*

Failure #4: No Ground Truth (Correctness)

AI cannot check the design against the **real domain**. Is the fare really per-minute? Does a rental really lock one bike? AI doesn't know — it guesses plausibly.

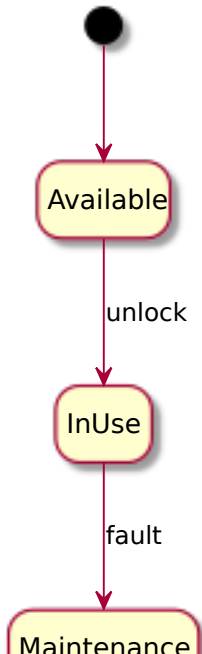
Lesson: *Correctness needs domain truth. Only you (and the stakeholder) have it.*

Failure #5: The Self-Evaluation Illusion

Ask the model that **generated** an artifact to grade it, and it rates its own work highly. There is no independent check — same blind spots, twice.

Lesson: *The generator can't be the only critic. That's the whole architect-and-critic split.*

Where AI Can Help



How to Use AI as an Evaluator

A fixed order, every time:

1. Ask **narrow, checkable** questions (conformance / consistency) — never “is this good?”.
2. Never let the **generator grade itself** — bring an independent check.
3. **Own completeness and correctness** yourself.
4. **Verify** every AI evaluation against the domain.

The Critique Loop, Inverted

ask a specific check -> verify AI's answer against the artifact ->
never accept "looks good".

When AI evaluates, **you** evaluate the evaluation. Same
architect-and-critic loop, one level up.

The Human Is the Evaluator of Record

AI can flag a missing multiplicity or a dead-end state. Judging whether the design is **complete** and **correct** — that's yours.

It's what the oral defense checks, and AI can't do it for you.

This Is Why the Course Drilled YOUR Critique

Every read-order — multiplicity, fabricated message, orphan state, broken trace — built the **human evaluator** that AI cannot replace.

The literacy floor is not nostalgia for UML. It is the judgement AI is worst at.

Next Week: Presentation & Defense Skills (Week 12)

You can evaluate. Next: how to **present and defend** an AI-mediated design.

Week 12: the Critique, Rationale, Traceability rubric explained, what examiners look for, dry-run mechanics. Plus **Lab 6**, your cold-defense dry run.

Critique, Rationale, Traceability — The Through-Line

Today you drilled:

- ▶ **Critique** — saw why your critique can't be delegated: AI evaluates to please, looks-right over is-right, blind to what's missing.
- ▶ **Traceability** — completeness and correctness across the trace are the human's to judge.

Rationale: when you reject AI's "looks good", you say *why* it isn't.

That's It For Today

- ▶ Next lecture (Week 12): presentation & defense skills.
- ▶ Looking ahead: Lab 6 (Week 12) is your cold-defense dry run.

Questions?